Practical Radiation Oncology (2016) 6, e39-e46

practical radiation oncology
pro

www.practicalradonc.org

Original Report

Enhancing safety and quality through preplanning peer review for patients undergoing stereotactic body radiation therapy

Martha M. Matuszak PhD [a,*], Scott W. Hadley PhD [a], Mary Feng MD [a], James A. Hayman MD [a], Kristy K. Brock PhD [a], Pamela Burger CMD [a], Dawn Owen MD, PhD [a], Krithika Suresh MMath [b], Matthew Schipper PhD [a,b], Theodore S. Lawrence MD, PhD [a], Jean M. Moran PhD [a]

[a]*Department of Radiation Oncology, University of Michigan, Ann Arbor, Michigan*
[b]*Department of Biostatistics, University of Michigan, Ann Arbor, Michigan*

Received 28 February 2015; revised 11 August 2015; accepted 17 September 2015

Abstract

Purpose: Because of its high dose per fraction delivery, stereotactic body radiation therapy (SBRT) requires real-time process assurance to promote safe, high-quality treatments. In an effort to assure safety and first-time quality, we instituted a pilot, single-institution, SBRT peer review process before treatment planning. Here, we present a summary of the results of that process over a 26-month period.

Methods and Materials: Before planning, all patients were presented at an SBRT "rounds" that required, at a minimum, the treating attending or resident physician, an independent attending physician, a physicist, and a dosimetrist. Items reviewed included imaging, image registration, target contours, prescription, and planning goals. The results of peer review were prospectively recorded and logistic regression models were used to assess the relationship between various physician and case characteristics and the odds of a change being made.

Results: A total of 513 SBRT cases were peer reviewed before planning. In 22.6% of cases, at least 1 change was made because of this process. A lower change rate was observed in higher volume SBRT body sites (lung and liver). In all body sites, gross and planning target volume contours were changed 8.2% and 5.5% of the time, respectively. The prescription was changed 4.9% of the time, and organs at risk goals were changed 7.2% of the time. The odds of having a change were significantly lower when the treating oncologist had more SBRT experience.

Conclusions: Preplanning peer review by an independent physician, physicist, and dosimetrist resulted in changes in nearly one-quarter of SBRT patients, potentially preventing suboptimal treatments. The odds of a change being required were decreased in higher volume body sites and when the treating oncologist was more experienced with SBRT, underscoring the potential importance of peer review in uncommon SBRT sites and at low-volume SBRT centers.

This work was presented at the Patient Safety II Scientific Session of the 56th Annual Meeting of the American Society for Radiation Oncology in San Francisco, California, September 14-17, 2014.

Conflicts of interest: None.

* Corresponding author. Room B2C432, SPC 5010, 1500 E. Medical Center Drive, Ann Arbor, MI 48109.
E-mail address: marthamm@med.umich.edu (M.M. Matuszak).

http://dx.doi.org/10.1016/j.prro.2015.09.009
1879-8500/

Introduction

Stereotactic body radiation therapy (SBRT) is a high-precision, high-dose radiation therapy treatment that is given in 1-5 fractions. Treatment planning for SBRT requires significant effort and resources because of the nature of the treatment. High plan conformity is required and target and normal tissue dose limits are strict to provide local control while minimizing the risk of toxicity. Because of the short time-course of treatment, traditional methods of peer review and quality assurance may not be optimally structured or timed. At our institution and many other radiation oncology departments, peer review for radiation therapy treatment plans is required and performed during a once-weekly chart rounds review.[1] As such, many SBRT plans reviewed in weekly chart rounds may already be completed or partially completed before review. Therefore, the threshold for making a change may be high and, in the cases in which a change is required, the overall quality of the treatment may be compromised because a potentially suboptimal plan could have already been partially delivered. In addition to the poor timing of peer review, observations in our clinic suggested that SBRT planning also suffered from inefficiency when incomplete instructions were provided to dosimetrists in terms of plan prescription, normal tissue dose limits, and presence of prior treatment requiring conservative adjustment of treatment planning goals.

In an effort to enhance safety and quality in the field, the American Society for Radiation Oncology (ASTRO) commissioned a series of reports to cover the safety and quality aspects of radiation oncology, including a report on enhancing the role of peer review in the field and another report on the quality and safety considerations for stereotactic treatments.[1,2] The 2013 report on peer review noted that it is one of the most effective means of quality assurance available to our field. However, the report also noted that this process is challenging and it would be impractical to peer review all aspects of plans. The report goes on to make many recommendations to aid in prioritization, timing, and workflow for peer review. In their prioritization, target volume definition was given the highest priority, followed by the decision to use radiation therapy, the planning directive, and the technical plan

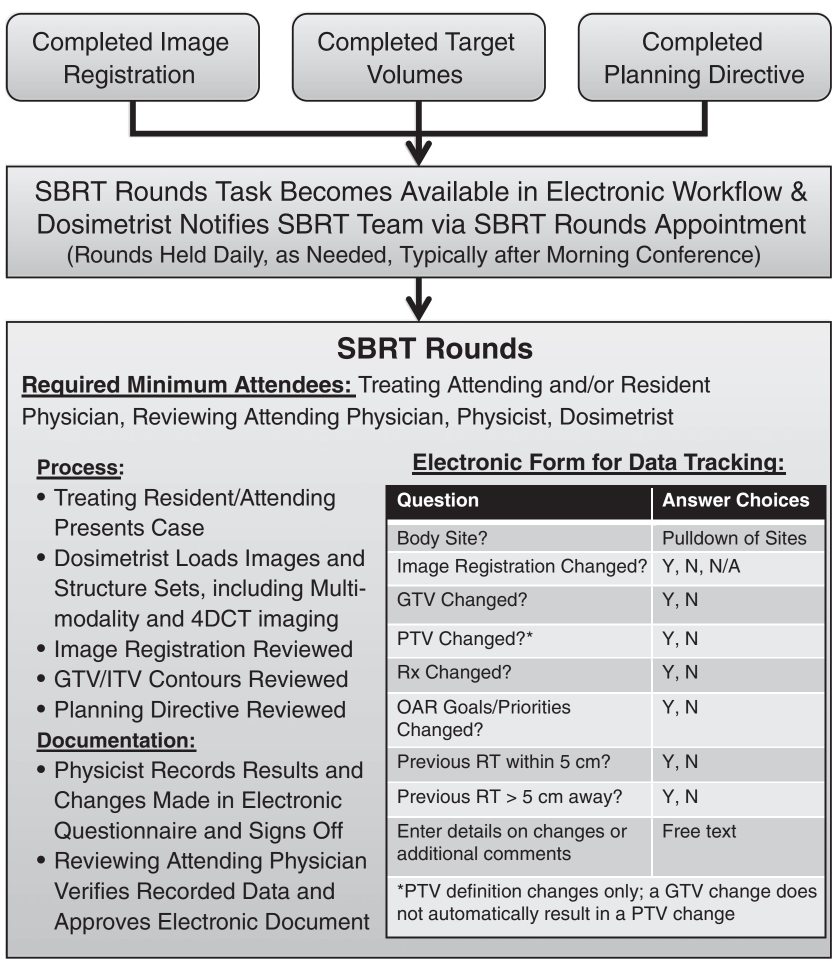

Figure 1 Basic workflow and electronic data tracking form involved with stereotactic body radiation therapy preplanning peer review.

quality and treatment delivery. The optimal timing for review of target volumes and the planning directive was suggested to be before treatment planning. The full report on quality and safety in stereotactic radiosurgery and SBRT recommended ongoing preplanning conferences (in addition to posttreatment review) for SBRT patients as an essential aspect when developing a new SBRT program.[2]

From our own experience, it was clear that postplanning weekly peer review, although valuable, was inadequate with respect to the detail and timeliness required for SBRT peer review. In addition, a 2009 investigation of failure modes in radiation oncology assigned 1 of the top 5 risk scores to using the incorrect contours in treatment planning, providing even more motivation for a strong peer review program before treatment planning.[3] Thus, in an effort to enhance safety and improve first-time quality for treatment plans, we instituted a single-institution, peer review pilot program before treatment planning for all SBRT patients. The goal of the review was to assess simulation imaging and registration, target contours, and the treatment planning directive to ensure upfront quality of the input data for SBRT planning. Here, we present a summary of the results of that process over a 26-month period.

Methods and materials

SBRT preplanning peer review process

As a pilot program in addition to weekly treatment planning chart rounds to review plan quality, all SBRT patients were presented at preplanning SBRT peer review rounds that required, at a minimum, the treating attending and/or resident radiation oncologist, an independent attending radiation oncologist, a physicist, and a dosimetrist. The basic workflow and process is shown in Fig 1. Before review, target volumes, image registration (if applicable), and a treatment planning directive were completed by the treating physician. The directives at our institution are customizable templates of instructions related to target and normal tissue definition, planning target volume (PTV) margins, dose prescription, normal tissue dose objectives and planning priorities, and information regarding image guidance instructions, previous treatment, and other planning alerts or instructions. When the “SBRT rounds” task became available in the patient’s electronic workflow, the dosimetrist would send out an appointment to the clinical SBRT team for rounds the following morning. Urgent cases were reviewed the same day.

To begin rounds, the resident and/or attending physician presented the case and relevant patient history, diagnosis, and plan for treatment. The dosimetrist displayed the imaging and contours in the treatment planning system. The gross tumor volume (GTV) was reviewed on the image it was defined on using multiple planes and views. If the GTV was contoured on a dataset that was not the primary planning dataset, the image registration was also reviewed. In addition, the contours were analyzed on the fused dataset for comprehensive review of multimodality target delineation. If an internal target volume (ITV) was drawn or created using 4-dimensional computed tomography (4D-CT) or multiple datasets, then the ITV was reviewed with respect to the source images in multiple planes. The results of peer review were prospectively recorded using a questionnaire in our oncology information system (Aria, Varian Medical Systems). We also tracked whether patients had received previous external beam radiation therapy. Fig 1 shows the questions on the electronic questionnaire that were used for tracking and data analysis.

A physicist recorded the result of rounds into the questionnaire and the questionnaire entries were queried and loaded into a document in the oncology information system. This document was reviewed for accuracy and approved by the peer review physician. Changes suggested during peer review were performed at the time of review unless additional review was required offline. Changes that were discussed but ultimately not implemented were not recorded as changes. In instances of disagreement, the treating physician was responsible for making the final decision.

Data extraction and analysis

Data from the tracking questionnaires was extracted from the treatment management system using a Structure Query Language statement to query the database. The data were evaluated for incomplete, duplicate, or unapproved questionnaires and those were excluded from the analysis. The results of peer review were analyzed to determine the number of cases that resulted in a change in any of the tracked criteria. For cases that required a change, details of the changes were recorded in the free-text comments. These comments were retrospectively coded by a physician and physicist to group types of changes and contributing factors to changes. In addition, the distribution of cases reviewed and rate of changes from each body site and the distribution of cases with previous radiation therapy were also calculated.

Univariate and multivariate analyses were performed to identify factors that affected the occurrence of a change made during SBRT peer review. Potential factors that were analyzed were the experience level of the treating radiation oncologist, the body site treated, and the patient’s previous radiation therapy status. The treating oncologist’s experience was considered both as a count of the oncologist’s previously reviewed SBRT cases and as an indicator of whether the oncologist had previously treated at least 5 SBRT cases (since the beginning of the tracking period). Logistic regression models were constructed to determine

Table 1 Distribution of body sites and change rate per body site for SBRT cases undergoing peer review

Body site	No. of cases peer reviewed	Percentage of total SBRT load	Cases with any change	Rate of change in body site
Liver	207	40.4	37	17.9
Lung	124	24.2	21	16.9
Brain	89	17.3	26	29.2
Other	36	7.0	14	38.9
Spine	35	6.8	10	28.6
Node	12	2.3	5	41.7
Kidney	10	1.9	3	30.0

SBRT, stereotactic body radiation therapy.

the significant predictors associated with a change being made as a result of SBRT preplanning peer review. To account for the cluster effect of patients being treated by the same radiation oncologist, parameter estimation was conducted using the Generalized Estimating Equations approach. Associations between the predictors and a change made are reported using odds ratios (OR) and their corresponding 95% confidence interval (95% CI). The statistical analyses were conducted using R 3.1.2 software (Revolution Analytics).

Results

Cases reviewed

From November 2012 to February 2015, 538 electronic questionnaires were recorded during peer review before SBRT treatment planning. Of those records, 25 were excluded from analysis because they were unapproved, incomplete, or duplicate entries. The 513 remaining records for patients undergoing SBRT of any body site were then analyzed to determine the impact of performing peer review before treatment planning. The data represent cases treated by 12 different radiation oncologists. Table 1 shows the distribution of body sites in the cases that were reviewed. Overall, 20.1% (103/513) of cases had previous treatment within 5 cm of the intended target, whereas 25.1% (129/513) had previous treatment more than 5 cm away from the intended target. There were 27 (5.3%) patients that returned for a second course of SBRT during the recording period.

Table 2 The types and frequency of changes made as a result of SBRT peer review

Type of change made	No. of changes	Percentage
Image registration	8	2.7 [a]
GTV	42	8.2
PTV	28	5.5
Prescription	25	4.9
OAR goals/priorities	37	7.2

GTV, gross tumor volume; OAR, organs at risk; PTV, planning target volume; SBRT, stereotactic body radiation therapy.

[a] Percentage of cases with an image registration.

Process and workflow

Overall, the SBRT peer review process was well-managed within the department. Based on a physician survey, the addition of SBRT preplanning peer review rarely delayed the time from simulation to treatment for patients; if there was a delay, it was never more than 1 day. The average time spent reviewing each case ranged from 5 to 20 minutes depending on the complexity of the case and subject of discussion. Cases with multimodality image registration, 4D-CT–based contours, and extensive prior treatment history required additional time. By comparison, approximately 2 minutes per case were spent on review of contours and treatment plans in our weekly chart rounds.

Analysis of changes made

At least 1 change was made as a result of preplanning peer review in 116 of the 513 cases reviewed (22.6%), and 22 cases (4.2%) had 2 or more changes. Table 1 shows the breakdown of changes as a function of body site treated. In body sites with the highest volume of SBRT treated (lung and liver), the overall change rate was observed to be the lowest. In 12 cases (2.3%), a change was discussed but not implemented by the treating attending physician.

Table 2 details the types and frequency of changes made overall. The most common change was made in the definition of the GTV with 8.2% (42/513) of cases requiring changes. The types of GTV changes observed based on the retrospective coding of free-text comments is shown in Table 3. Fig 2 shows an ITV change made after reviewing the tumor motion extent on 4D-CT near the diaphragm. Because of uncertainty in the volume, a substantial portion of volume was added to the ITV. Fig 3 shows an example of a patient in whom the lung GTV was mistakenly contoured on an exhale contrast scan (acquired for a simultaneous abdominal SBRT treatment) instead of the 4D-CT. The final, corrected ITV was 5 mm larger because of the inclusion of motion.

The second most common change was to the goals or priority level of the organs at risk (OAR), with changes being made in 7.2% (37/513) of cases. Based on the retrospective coding of changes, it was determined that at least 11 of the 37 changes made to the OAR goals resulted from consideration of previous treatment. Other OAR changes are listed in Table 3.

Changes to the PTV were made in 5.5% (28/513) of cases. Reasons contributing to changes are given in

Table 3 Retrospective scoring of types of changes and contributing factors

Type of GTV change	No. of changes attributed[a]	% of changes attributed[b]
Minor edits on 1-2 axial cuts	12	42.9
Moderate increase in GTV size or extent	7	25.0
Edited after review of additional imaging	3	10.7
Edited after review of 4D-CT motion	3	10.7
Added a CTV because of uncertainty in GTV	2	7.1
Additional GTV added	1	3.6
Undetermined	14	N/A

[a] More than 1 type of change could be scored per case. The total number of cases with GTV changes was 42/513.
[b] Percentage of cases with determinable GTV changes (28).

Contributing factor to OAR change	No. of changes attributed[a]	% of changes attributed[b]
OAR or objective missed on original directive	12	41.4
Previous RT	11	37.9
Removed an objective that was too conservative	3	10.3
Changed priority vs. target	2	6.9
Added a PRV	2	6.9
Undetermined	8	N/A

[a] More than 1 contributing factor could be scored per change. The total number of OAR changes was 37/513.
[b] Percentage of cases with determinable OAR changes (29).

Type of PTV change	No. of changes attributed[a]	% of changes attributed[b]
Added margin because of registration uncertainty	6	40.0
Changed margin resulting from lack of standard definition for site	4	26.7
Added margin from motion uncertainty	2	13.3
Added margin to account for IGRT uncertainty of treating multiple GTVs in 1 plan	2	13.3
Fixed PTV because of stray pixels	1	6.7
Undetermined	13	N/A

[a] More than 1 type of change could be scored per case. The total number of cases with PTV changes was 28/513.
[b] Percentage of cases with determinable PTV changes (15).

Contributing factor to prescription change	No. of changes attributed[a]	% of changes attributed[b]
Previous RT	10	45.5
OAR toxicity concerns	6	27.3
More aggressive treatment suggested	6	27.3
Lack of standard prescription for site	5	22.7
Change to non-SBRT prescription	4	18.2
Prescription did not follow standards	1	4.5
Undetermined	3	N/A

[a] More than 1 contributing factor could be scored per change. The total number of prescription changes was 25/513.
[b] Percentage of cases with determinable prescription changes (22).

4D, 4-dimensional; CT, computed tomography; CTV, clinical tumor volume; GTV, gross tumor volume; IGRT, image guided radiation therapy; OAR, organs at risk; PRV, planning risk volume; PTV, planning target volume; SBRT, stereotactic body radiation therapy.

Table 3; however, nearly half of the PTV changes could not be retrospectively coded because of a lack of detail in the free-text comments. When determinable, PTVs were most commonly increased because of image registration uncertainties.

Changes to the prescription (25/513 or 4.9%) and image registration (2.7% of cases with a defined registration) were the least common changes recorded. Specific reasons scored as contributing to a change are listed in Table 3. Of the 25 cases with a prescription change, 10 were scored as resulting from previous radiation therapy considerations. In 4 cases, the patients were switched to a non-SBRT fractionation scheme. Of the other cases with prescription changes, 5 were in body sites (abdominal nodes, spine, and adrenal gland) that, at the time, did not have standardized dose prescriptions for SBRT in our clinic.

Statistical analysis

Table 4 shows the ORs and 95% CIs from the models fit in univariate and multivariate analyses that revealed similar associations. Patients with a primary site of lung or liver cancer had a decreased probability of having a change compared with sites with a smaller volume of cases, but only cases classified as kidney, node, or other had significantly increased odds of a change compared with lung cancer cases (OR, 2.84; 95% CI, 1.22-6.60) and liver cancer cases (OR, 2.22; 95% CI, 1.32-3.75). Previous treatment more than 5 cm away from the target was associated with significantly increased odds of a change (OR, 2.49; 95% CI, 1.46-4.24) and previous treatment within 5 cm of the target resulted in increased odds of an OAR change (OR, 2.42; 95% CI, 0.92-6.32). For patients treated by a radiation oncologist with 5 or more previous SBRT cases reviewed, the odds of a change was significantly lower (OR, 0.51; 95% CI, 0.28-0.93). The odds of a patient's SBRT plan needing a change decreased by a factor of 0.984 (95% CI, 0.970-0.997) for every additional 5 patients treated by their oncologist.

Qualitative observations

In addition to highlighting changes, the questionnaire comments recorded common points of discussion during rounds. In general, the comments revealed that SBRT rounds prompted discussion on the original indication for SBRT, mitigation from previous radiation therapy, the efficacy of the selected prescription dose and fractionation, treatment planning and image guidance strategy, and the future treatment of additional lesions. We have also observed that changes have been made to treatment planning directive templates as a result of discussion in rounds in an effort to improve the clarity of instructions given to planners and standardization between body sites.

Discussion

Our SBRT peer review procedure is in line with ASTRO guidance on peer review, which suggests that target definition and planning directive review are level 1 and 2 priorities, respectively.[4] In fact, peer review target segmentation was thought to be "one of the most important medical decisions that would benefit from peer review" and of especially high impact for SBRT. We saw GTV changes in approximately 8% of all cases reviewed. Because the clinical impact of GTV changes can vary, our updated procedure for SBRT review will incorporate physician grading of potential clinical impact of different interventions.

Perhaps the 2 most interesting findings of our work were a reduction in the rate of changes resulting from peer review in body sites with a high volume of cases and decreased odds of a change being made in cases treated by SBRT-experienced physicians. This points to the importance of peer review in smaller volume centers and the

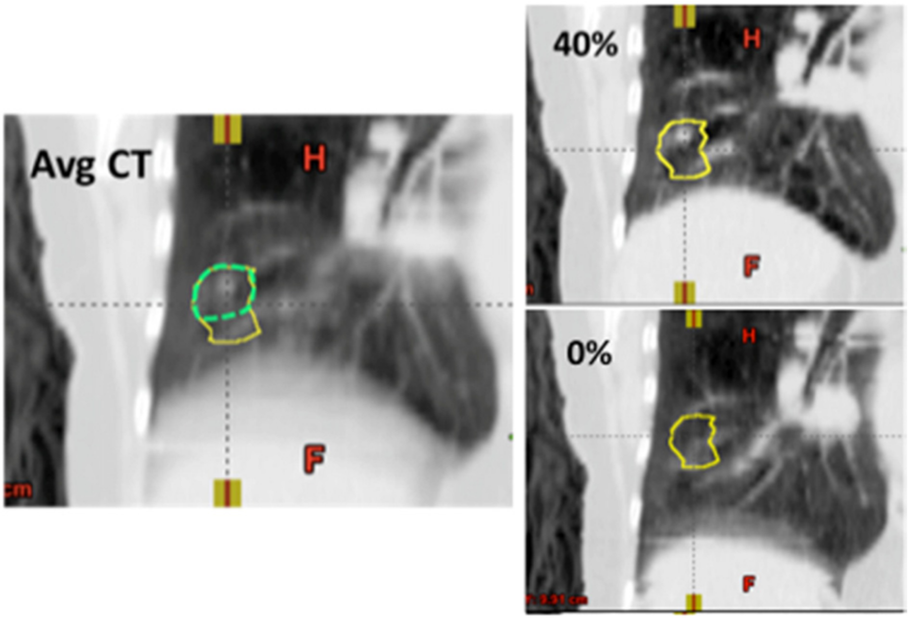

Figure 2 Example of a patient in whom the internal target volume (ITV) was adjusted (solid contour) to include additional volume noted on the 0% phase of the 4-dimensional CT (lower right) compared with the simulated originally drawn ITV (dashed contour), which did not include the inferior aspect of the ITV. Avg, average; CT, computed tomography.

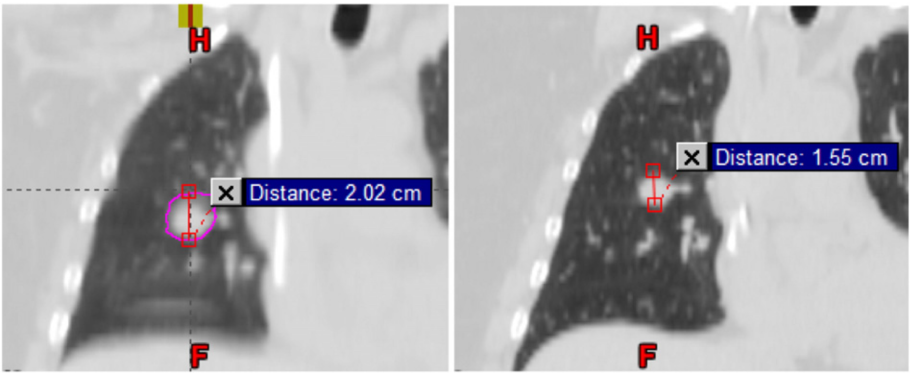

Figure 3 Example of a patient where the gross tumor volume was redefined because of mistakenly being contoured on a breath-hold contrast scan (right) taken for an abdominal stereotactic body radiation therapy course instead of creating an internal target volume on the average 4-dimensional CT scan (left).

potential for peer review partnerships with larger networks. In addition, the reduction in changes as the physician gains experience follows the hypothesis proven in other studies that there is a learning curve associated with advanced techniques.[5,6] Still, the high rate of overall changes supports the idea that peer review in this setting is extremely valuable, even for experienced centers and physicians.

It is not surprising that patients with previous radiation therapy have a higher chance of a change being made after peer review. Although having previous treatment close to the current target was not a significant predictor of any change, it was a predictor of a potential OAR change. In fact, an early area we addressed from SBRT peer review was to improve the handling of cases with previous treatment prior to going through SBRT rounds. Although changes still occur, only 2 of the 11 changes in OAR goals that were attributed to prior radiation therapy concerns occurred in the most recent 12 months of tracking data. Changes made because of prior radiation have the potential to be high impact and highlight the safety aspects of a detailed preplanning review. Although we advocate preplanning review on all SBRT cases, for institutions that are resource-limited, a focus on nonstandard cases and cases with prior radiation therapy is recommended.

Brundage et al reported on a pretreatment audit procedure for more than 3000 plans over 8 years.[7] In their review, changes were suggested in 7.7% of plans, with 4.1% resulting from problems identified in the target volumes, critical structures, or dose distribution. Of these, the target volume changes were the most common. However, the majority of those plans were never corrected, but used as feedback to improve future quality. The latter suggests 1 of the benefits of preplanning review is that most changes are applied because the threshold for making changes is much lower than postplanning initiation.

There are several limitations of this work. First, it is difficult to determine the increase in efficiency or time saved in replanning because of SBRT preplanning review. To say that all 116 cases that required at least 1 change because of SBRT rounds were saved from a replan would not be accurate. Some percentage of changes may have been identified during the planning process, physics chart review, or weekly postplanning chart rounds, but it is unknown if they would have been acted upon. We believe that the overall quality of our SBRT program has improved as a result of this peer review process, but it is difficult to show quantitatively. Another limitation of our work is that we did not track the potential clinical impact of the changes made. An important aspect of future research into peer review is availability of prospective grading of the potential clinical impact of changes. Finally, the retrospective scoring of free-text comments to determine the contributing factors to changes was not ideal because many changes were undetermined. Improved tracking of the types of changes and contributing factors will be applied in the future.

With a busy academic clinic, it does remain a challenge to ensure the required individuals are available for peer review on a daily basis. We have found it important to be flexible with scheduling and for involved parties to make SBRT review a priority so as to not delay treatment planning. Another positive aspect around SBRT peer review has been a department-wide shift in the culture around peer review. The atmosphere around SBRT rounds is very respectful and nonpunitive, with many physicians using it as an opportunity to discuss challenging cases with their peers and for education of the residents.

It is worth mentioning that the electronic workflow and reporting tools implemented as part of this initiative were successful and enabled timely review of cases. The management and integration of peer review within standard workflow was noted as an important factor in a strong program in the 2013 ASTRO report on peer review.[4]

Finally, we should note that, although this process started as a pilot program, it was quickly integrated into our standard workflow because of the value found in the process. We continue to review all SBRT patients before

Table 4 Results of univariate and multivariate statistical analyses

Parameter	Univariate analyses			Multivariate analyses		
	OR	95% CI	*P* value	OR	95% CI	*P* value
Body site (reference = lung)						
Liver	1.07	(0.54-2.11)	.85	1.28	(0.70-2.32)	.43
Spine	1.96	(1.14-3.37)	.01	1.77	(0.80-3.87)	.16
Brain	2.02	(0.87-4.72)	.10	2.04	(0.92-4.52)	.08
Kidney, node, or other	3.00	(1.37-6.58)	.01	2.84	(1.22-6.60)	.02
Previous RT within 5 cm of target site	1.55	(0.65-3.66)	.32	1.31	(0.49-3.46)	.59
Previous RT >5 cm away from target	2.49	(1.51-4.10)	<.001	2.49	(1.46-4.24)	<.001
Treated by a physician with 5 or more SBRT cases previously reviewed	0.51	(0.26-0.99)	.05	0.51 [a]	(0.28-0.93)	.03
Number of previous SBRT cases reviewed by treating oncologist (per each additional 5 cases reviewed)	0.981	(0.966-0.996)	.02	0.984 [1]	(0.970-0.997)	.02

CI, confidence interval; OR, odds ratio; RT, radiation therapy; SBRT, stereotactic body RT.

[a] These estimates come from separate regression model fits that differ in the way that physician experience was coded.

planning and refine our process to meet the evolving needs of the clinic. The future of peer review for our SBRT program may include expanding SBRT rounds to be required at all affiliated community centers (we currently perform SBRT peer review, when requested, for our community practices). In addition, we are currently determining the need to include postplanning, pretreatment peer review in addition to preplanning review. At the current time, all SBRT treatment plans are still reviewed in weekly chart rounds.

Conclusions

Preplanning peer review for SBRT had a positive effect on quality of treatment volumes, treatment prescription, and planning priorities. In our experience, changes were made an impressive 22.6% of the time because of upfront review by an independent physician, physicist and dosimetrist. We observed a higher rate of changes for body sites with a lower volume of cases treated, suggesting the importance of peer review in nonstandard cases. We also observed that physician experience with SBRT and the patient's previous treatment status were significant factors in whether or not a change was made due to peer review. Qualitatively, we believe that pretreatment planning review has improved both quality and safety by preventing potentially suboptimal treatments in nearly one-quarter of SBRT patients treated.

Acknowledgments

The authors acknowledge the team of physicians, physicists, and dosimetrists who routinely participate in peer review for stereotactic body radiation therapy as well as Xiaoping Chen for his assistance with data extraction for analysis.

References

1. Lawrence YR, Whiton MA, Symon Z, et al. Quality assurance peer review chart rounds in 2011: A survey of academic institutions in the United States. *Int J Radiat Oncol Biol Phys.* 2012;84:590-595.
2. Solberg TD, Balter JM, Benedict SH, et al. Quality and safety considerations in stereotactic radiosurgery and stereotactic body radiation therapy: Full report. *Pract Radiat Oncol.* 2012;2:2-9.
3. Ford EC, Gaudette R, Myers L, et al. Evaluation of safety in a radiation oncology setting using failure mode and effects analysis. *Int J Radiat Oncol Biol Phys.* 2009;74:852-858.
4. Marks LB, Adams RD, Pawlicki T, et al. Enhancing the role of case-oriented peer review to improve quality and safety in radiation oncology: Executive summary. *Pract Radiat Oncol.* 2013;3:149-156.
5. Huang G, Medlam G, Lee J, et al. Error in the delivery of radiation therapy: Results of a quality assurance review. *Int J Radiat Oncol Biol Phys.* 2005;61:1590-1595.
6. Marks LB, Light KL, Hubbs JL, et al. The impact of advanced technologies on treatment deviations in radiation treatment delivery. *Int J Radiat Oncol Biol Phys.* 2007;69:1579-1586.
7. Brundage MD, Dixon PF, Mackillop WJ, et al. A real-time audit of radiation therapy in a regional cancer center. *Int J Radiat Oncol Biol Phys.* 1999;43:115-124.